Optimal Scheduling of Battery Energy Storage System Performing Stacked Services

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***Abstract*—Grid-scale battery energy storage systems (BESSs) are at the forefront of technologies utilized to provide stability and flexibility to the power grid. As a result, BESSs generate significant revenue for their operators by participating in ancillary services such as the energy arbitrage and frequency regulation markets. Therefore, BESS operators can benefit from a model that allows them to optimize the bidding process for providing services while also optimizing scheduling in a way that best exploits each BESS cycle by simultaneously stacking various grid services. Estimating maximum BESS revenue is crucial to establishing financial sustainability for investors. In this paper, a BESS optimization model for multiple grid applications is proposed to estimate maximum daily revenue with an appropriate focus on maintaining the longevity of the BESS. The model aims to maximize the revenue generated by a BESS by allowing the system to participate in the energy arbitrage and frequency regulation markets at the same time. In this proposal, a new BESS scheduling method, tested using historical PJM market data, is used to improve revenue generation from providing ancillary services through effective and optimized bidding into the PJM regulation market based on buying/selling to/from the energy markets. The model is formulated as a mixed-integer linear programming (MILP).**

***Index Terms*—battery energy storage systems (BESSs), frequency regulation up/down market, ancillary services, energy arbitrage, bidding capacity, scheduling optimization, BESS cycles.**

NOMENCLATURE

Indices

d	Index for day.
t	Index for inter-hour time periods [1,24].
k	Index for intra-hour time periods [5-min].

Parameters

$R\mathrm{MCCP}_{ktd}$	Regulation market capability clearing price.
$R\mathrm{MPCP}_{ktd}$	Regulation market performance clearing price.
S_{td}	BESS actual performance.
β_{ktd}	Mileage ratio.
δ_{td}	Real-time electricity price in \$/MWh.
E^R	BESS energy rating in MWh.
P^R	BESS power rating in MW.
$R\mathrm{egD}^{up}$	Ramp-up AGCS signal (per unit).
$R\mathrm{egD}^{down}$	Ramp-down AGCS signal (per unit).

Variables

C	Number of cycles performed by BESS.
C_{max}	Maximum number of BESS daily cycles.
P_{td}^{EAd}	BESS discharged power for energy arbitrage.
P_{td}^{EAc}	BESS charged power for energy arbitrage.
P_{ktd}^{reg}	BESS available capacity for frequency regulation.
P_{ktd}^{rdb}	Ramp-down available capacity for bidding in MW.
P_{ktd}^{rub}	Ramp-up available capacity for bidding in MW.
P_{ktd}^{rda}	Actual ramp-down power used for frequency regulation depending on AGC signal.
P_{ktd}^{rua}	Actual ramp-up power used for frequency regulation depending on AGC signal.
J_{ktd}	Total available power that can be used for frequency regulation at time interval.
u_{td}^1	Binary variable associated with BESS discharging (selling) for energy arbitrage.
u_{td}^2	Binary variable associated with BESS charging (buying) for energy arbitrage.
u_{ktd}^3	Binary variable associated with the participation of the BESS in frequency regulation market.
z_{ktd}^{rua}	Binary variable associated with BESS actual frequency regulation up.
z_{ktd}^{rda}	Binary variable associated with BESS actual frequency regulation down.
E_{ktd}^{soc}	Energy stored in the BESS at each time interval.

I. INTRODUCTION

Due to the high rate of growth in integrated renewable energy and the increase in power consumption of the U.S. population, maintaining the stability of the power grid is a critical challenge, one that requires resources with fast ramping rates that can quickly meet need. One such resource is that provided by third-party battery energy storage systems (BESSs), which serve the power grid in a variety of ways that keep it stable and flexible, particularly during peak hours [1]. Given the fast ramping rate of BESSs, they can quickly be called upon to act as a load or power source, making them suitable for both frequency regulation and voltage support. Furthermore, utilization of third-party BESSs allows Independent System Operators (ISOs) and Regional Transmission Organizations (RTOs: hereafter collectively referred to as RTOs) to maintain power grid frequency without costly equipment upgrades and other infrastructure expenses [2]. The North American Electric Reliability Council (NERC) mandates the reliability criteria that the U.S. power grid must meet at all times to ensure generation is consistent with demand. In case of frequency changes, RTOs must act immediately to prevent blackouts. A BESS can support frequency in both up or down scenarios. When generation suddenly increases, a BESS can be charged and act as a load. Conversely, when demand is higher, a BESS can act as a generator and discharge its power to the power grid. As a result of this flexibility and ease of utility, BESS investment in the ancillary services market has been rapidly expanding, particularly over the course of the roughly 10 years since the Federal Energy Regulatory Commission (FERC) issued FERC Order 755 in 2011 and then expanded upon that order's pay-for-performance requirements with FERC Order 784 in 2013. These orders introduced a new compensation structure that requires RTOs to compensate frequency regulation service providers not simply for the capacity they provide but also for their performance in providing that capacity, a change that benefited BESSs [3]. In this case, performance is related to ramp-up speed, thereby creating an incentive for fast-ramping resources such as BESSs. Therefore, these compensation practices have created a better climate for BESSs and have made participating in the ancillary services markets more appealing and profitable to BESS operators [4]. However, the accompanying increase in ancillary service providers in these markets requires BESS operators employ innovative practices to remain competitive and profitable.

In the last decade, RTOs in the United States including PJM, CAISO, and ERCOT have adopted a new payment mechanism for performance that includes mileage ratios and performance scores [5]. Before the issuance of FERC Order 755, the PJM ancillary services market only compensated regulation resources based on their capacity, regardless of how fast regulation resources could follow the signal sent by RTOs. The accuracy and deployment of regulation resources were ignored.

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One of the earliest economic analyses of the applications of BESS bulk energy was conducted in 2012 by Sandia National Laboratories [1]. This report presented a comprehensive study of the potential benefits of grid-scale BESSs as well as well as the value of participating in both the energy arbitrage and frequency regulation markets for a flywheel BESS. In [6], a state of charge control model was proposed to maximize frequency regulation market profits by minimizing operation costs rather than improving bidding and scheduling methods. A profit maximization model that incorporated energy arbitrage and frequency regulation was proposed in [7] to evaluate the value of BESS investment in the European context. In this study, significant technical constraints, such as depth of discharge and life cycles, were ignored. In [8], a 1 hr time resolution was used for three individual BESS services (i.e., energy arbitrage, frequency regulation, and outage mitigation) to maximize BESS profitability. Although the study considered a long-term investment, the BESS's lifetime was not considered. The work in [9] investigated the economics of BESSs in the PJM market. To do so, the authors developed a bidding strategy for frequency regulation to avoid dropping market prices and retain profits, using only a single BESS application.

BESSs can be used in several applications simultaneously and still earn a profit. Previous research on the economics of BESSs in terms of grid applications has implied that frequency regulation is the most profitable application for BESS investors/operators. This neglects to consider the fact that providing ancillary services–especially frequency regulation–requires fast response and a high number of daily cycles. These factors decrease the overall lifetime of the BESS and therefore significantly affect profit in the long run. Thus, when developing a model for stacked applications, several factors should be considered, including bidding strategies, depth of discharge, and number of BESS daily cycles required. To our knowledge, none of the previous research has discussed a BESS performing stacked applications simultaneously or the application of inter- and intra-hour scheduling. To address this gap in the research, we developed a model that can benefit from each BESS's cycle. Our proposed model allows the BESS to operate in both markets simultaneously, when possible. This methodology generates significantly greater revenue in a more efficient manner without negatively impacting the longevity of the BESS. The optimization model applies inter- and intra-hour scheduling for the BESS and several binary variables to ensure BESS operations. The remainder of the paper is organized as follows. Section II describes and explains the grid applications used in this work. Section III presents the mathematical models of the optimization problem used to calculate revenue from the grid applications. Next, Section IV describes the parameters of our BESS case study and presents the results. Section V is the conclusion and also describes future work.

II. Grid Application of BESSs

A. *Energy Arbitrage*

In energy arbitrage, the BESS is charged during off-peak hours when energy costs are low and then, alternatively, that stored energy is used during peak demand periods for other applications, such as frequency regulation up, peak shaving, and voltage support. During the process of energy arbitrage, locational marginal pricing (LMP), an indicator for wholesale energy prices at a particular location at a specific time, is used to determine when to charge and discharge the BESS. In this way, BESSs take advantage of price and generation variation in the grid, especially when renewable energy resources are integrated or during low-demand periods.

B. Frequency Regulation

Variable renewable power generation and load fluctuations continuously increase system frequency fluctuations. Thus, generation assets, such as thermal units, are ramped to improve regulation. This action requires high efficiency and frequent ramping up or down, which results in high thermal unit fuel consumption. Therefore, considering using BESSs for frequency regulation is more reliable and efficient than other methods [10].

In the frequency regulation market, BESSs bid capacity, either up or down, into the market. Bidding capacity up means that the BESS can discharge capacity into the power grid at a specific time. When bidding down, the BESS has the capacity to be charged (consume capacity and act as a load) at a specific time. After this is done, the RTO sends a signal to specify the value of the bid capacity needed to maintain the grid frequency either up or down. BESSs get paid for what they reserved during bidding, even if the signal is zero. For frequency regulation, the PJM regulation market is selected for this study as it is typical and because of the accessibility of its historical market data [11].

III. Novel Mathematical Model for BESS Scheduling

In this section, first, an overview of the PJM pay-for-performance market mechanism is presented to summarize how the PJM pays frequency regulation market participants. Then, the BESS mathematical model for calculating revenue and the operation model for bidding into the frequency regulation market and participating in the energy market are explained.

A. PJM Market Mechanism

Based on FERC Order 755, the PJM developed a comprehensive pay-for-performance formula for power resources wanting to perform frequency regulation. This method considers several significant factors, such as the speed and accuracy of power resources along with the distance of power sources. The pay-for-performance credit considers how quickly and accurately a power source can track the signal provided by PJM. The PJM market has two different Automatic Generation Control Signals (AGCSs): RegD and RegA. RegD is developed and sent to fast-ramping resources with limited energy capacities, such as BESSs and gas turbines; RegA is for slow-ramping units, such as hydro units. The pay-for-performance credit is a combination of two payments:

1) Capability Payment: Based on regulation market capability clearing prices (RMCCPs) and only employs performance scores (*S*). The performance score of a resource is representative of how good it is at following AGCSs and depends on three components: accuracy, delay, and precision. In the PMJ, the scores are calculated at the end of each hour. A more detailed method of how to calculate these components is available at [11].

2) Performance Payment: Based on regulation market performance clearing prices (RMPCPs). In addition to the performance score, the performance payment includes a mileage ratio that measures the movement of different signals used to regulate frequency. For the PJM market, the mileage ratio is the ratio between RegA and RegD [11]. In 2018, the weighted average price of the RMPCP in the PJM market was $33 per megawatt, which was an increase of 118% over the same period in 2017. The RMCCP and RMPCP for the PJM market during the first week of September 2018 are shown in Figure 1. [12].

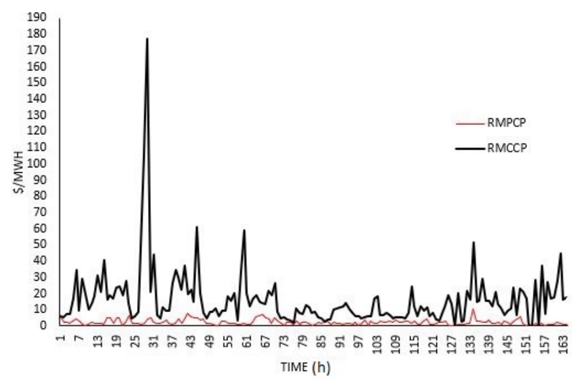

Fig. 1. Regulation prices for one week in the PJM market [12]

B. BESS Optimization Model

The proposed model for estimating revenue from the energy arbitrage and frequency regulation markets is defined in equation (1). The total revenue is the summation of power sold and bought in the energy market and the power bid (reserved) into the frequency regulation market. The first part of the equation is the total power in energy arbitrage multiplied by the electricity price represented by the LMP. The frequency regulations payments RMCCP and RMPCP are divided by *K* to represent a 5-min interval, according to the PJM manual [11].

$$\max \Bigg[\sum_{t} \sum_{d} \tau (P_{td}^{EAd} + P_{td}^{EAc})\ \delta_{td} + \sum_{k} \sum_{t} \sum_{d} P_{ktd}^{reg}\ S_{ktd}\ \frac{RMCCP_{ktd}}{K} + \sum_{k} \sum_{t} \sum_{d} P_{ktd}^{reg}\ S_{ktd}\ \beta_{ktd}\ \frac{RMPCP_{ktd}}{K} \Bigg] \quad (1)$$

The power bought from the energy market (charged into the BESS) and the power sold to the power grid (discharged from the BESS) are subjected to limitations by equations (2) and (3). Binary variables are used to make sure that the battery is not charging and discharging simultaneously equation(4).

$$0 \leq \; P^{EAd}_{td} \leq \; P^R \; u^1_{td} \qquad \forall t, \forall d \tag{2}$$
$$-P^R \; u^2_{td} \leq \; P^{EAc}_{td} \leq 0 \qquad \forall t, \forall d \tag{3}$$
$$u^1_{td} + u^2_{td} \leq 1 \qquad \forall t, \forall d \tag{4}$$

Frequency regulation reserved power consists of two parts: frequency regulation bidding up, where the BESS is offering a capacity to be discharged into the power grid to meet demand, and frequency regulation down, where the BESS is offering a capacity to be charged. The total power available to bid in the market for frequency regulation is calculated by equation (5).

$$P^{reg}_{ktd} = P^{rub}_{ktd} + P^{rdb}_{ktd} \qquad \forall k, \forall t, \forall d \tag{5}$$

Equation (6) is used to calculate the capacity of frequency regulation up values at the first interval of each starting hour.

$$0 \leq \; P^{rub}_{ktd} \leq (P^R - P^{EAd}_{td}) \; u^1_{td} \; u^3_{ktd} + P^R(1-u^1_{td}) \; (1-u^2_{td}) \; u^3_{ktd} \qquad \forall k=1, \forall t, \forall d \tag{6}$$

When k is greater than 1, equation (7) is applied for frequency regulation up.

$$0 \leq \; P^{rub}_{ktd} \leq (P^R - P^{EAd}_{td} - J_{(k-1)td}) \; u^1_{td} \; u^3_{ktd} + [(P^R - J_{(k-1)td}(1-u^1_{td})(1-u^2_{td})] \; u^3_{ktd} - J_{(k-1)td} \; u^2_{td} \; u^3_{ktd} \qquad \forall k \neq 1, \forall t, \forall d \tag{7}$$

Frequency regulation down is a negative value represented by equations (8) and (9) for the first interval of each starting hour and for the rest of the day, respectively.

$$-(P^R + P^{EAc}_{td})u^2_{td} \; u^3_{ktd} - (P^R)(1-u^1_{td})(1-u^2_{td}) \; u^3_{ktd} \leq \; P^{rdb}_{ktd} \leq 0 \qquad \forall k=1, \forall t, \forall d \tag{8}$$

$$-J_{(k-1)td} \; u^1_{td} \; u^3_{ktd} - (P^R + P^{EAc}_{td} + J_{(k-1)td}) \; u^2_{td} \; u^3_{ktd} - (P^R + J_{(k-1)td})(1-u^1_{td})(1-u^2_{td}) \; u^3_{ktd} \leq \; P^{rdb}_{ktd} \leq 0 \qquad \forall k \neq 1, \forall t, \forall d \tag{9}$$

The total actual power that is identified and charged or discharged from a BESS after bidding and receiving a signal is represented by equations (10) and (11), respectively.

$$J_{ktd} = P^{rua}_{ktd} + P^{rda}_{ktd} \qquad \forall k=1, \forall t, \forall d \tag{10}$$
$$J_{ktd} = (J_{(k-1)td} + P^{rua}_{ktd} + P^{rda}_{ktd}) \qquad \forall k \neq 1, \forall t, \forall d \tag{11}$$

The RegD signal in this work is divided into two signals: RegD$^{\text{down}}$ (between -1 and zero) and RegD$^{\text{up}}$ (between zero and 1). The negative indicates that the frequency in the power grid is high and the BESS should be charged to regulate the frequency down. The positive value is used to regulate the frequency up. If the signal is zero, that means the system is stable and no frequency regulation is needed. RegD is used to calculate the actual power that is used by bidding for frequency regulation up and down. equations (12) and (13) show the calculation methods for actual frequency regulation values using RegD$^{\text{up}}$ and RegD$^{\text{down}}$, respectively.

$$P^{rua}_{ktd} = P^{rub}_{ktd} \; RegD^{up} \; z^{rua}_{ktd} \qquad \forall k, \forall t, \forall d \tag{12}$$
$$P^{rda}_{ktd} = -P^{rdb}_{ktd} \; RegD^{down} \; z^{rda}_{ktd} \qquad \forall k, \forall t, \forall d \tag{13}$$

To ensure that the BESS is not charging and discharging simultaneously for the first interval of the day, equations (14) and (15) are applied.

$$u^1_{td} + z^{rda}_{ktd} \leq 1 \qquad \forall k=1, \forall t, \forall d \tag{14}$$
$$u^2_{td} + z^{rua}_{ktd} \leq 1 \qquad \forall k=1, \forall t, \forall d \tag{15}$$

Equations (16)-(18) are applied to ensure that the BESS can read the RegD signal and follow it when bidding. The binary variable z^{rda} is equal to 1 when RegD is less then zero. Similarly, z^{rua} is equal to 1 when RegD is greater than zero.

$$z^{rua}_{ktd} + z^{rda}_{kt} = u^3_{ktd} \qquad \forall k, \forall t, \forall d \tag{16}$$
$$u^3_{ktd} \; RegD^{up} \leq \; z^{rua}_{ktd} \leq 1 \qquad \forall k, \forall t, \forall d \tag{17}$$
$$-u^3_{ktd} \; RegD^{down} \leq \; z^{rda}_{ktd} \leq 1 \qquad \forall k, \forall t, \forall d \tag{18}$$

The state of charge is the energy stored in the BESS at each time interval and is calculated using equations (19) and (20). The state of charge is also subject to equation (21), where the BESS can be fully charged to its rated capacity and can discharge to the full depth of discharging limits D. It is important to note that the power in energy arbitrage for both cases (charging/discharging) and the actual capacity of power for frequency regulation (up/down) are multiplied by the time interval τ in (19)-(20) to be converted to energy and then added to the BESS state of charge.

$$E^{BES}_{ktd} = E^{BES}_{(k-1)td} - (\frac{P^{EAd}_{td} + P^{EAc}_{td}}{K})\tau + J_{ktd} \; \tau \qquad \forall k > 1, \forall t, \forall d \tag{19}$$

$$E^{BES}_{ktd} = E^{BES}_{Kd(t-1)} - (\frac{P^{EAd}_{td} + P^{EAc}_{td}}{K})\tau + J_{ktd} \; \tau \qquad \forall k=1, \forall t > 1, \forall d \tag{20}$$

$$(1-D)\, E^R \leq \; E^{BES}_{ktd} \leq \; E^R \qquad \forall k, \forall t, \forall d \tag{21}$$

To ensure that the BESS is able to perform several services, at the beginning of each day, equation (22) is applied to set the state of charge of the BESS at 50% of its rated capacity. This way, the BESS would have enough capacity for both up/down frequency regulation or energy arbitrage.

$$E^{BES}_{ktd} = 0.5 * \ E^{R} \qquad \forall k = 1, \forall t = 1 \tag{22}$$

The following equation is added to calculate the number of cycles and ensure the BESS will not exceed its limits; in addition, it also supports prolonging the BESS's lifetime for long-term projects. This method of calculating the number of cycles depends only on discharging. The third part of equation (23) is added to prevent calculating double-discharging cycles when the BESS is discharging in both markets simultaneously.

$$C_{ktd} = (u^1_{td} - u^1_{(t-1)d})u^1_{td} + (z^{rua}_{ktd} - z^{rua}_{k(t-1)d})z^{rua}_{ktd} - z^{rua}_{ktd}\ u^1_{td} \qquad \forall k, \forall t, \forall d \tag{23}$$

The maximum number of cycles can be limited by the following constraint.

$$\sum_{k}\sum_{t}\sum_{d} C_{ktd} \leq C_{max} \qquad \forall k, \forall t, \forall d \tag{24}$$

IV. CASE STUDY AND RESULTS

Two different scenarios are applied to this simulation to demonstrate how the proposed model affects the operation and scheduling of a BESS. We apply both applications simultaneously and then individually using the same time period with the same set of data from both markets. A Li-ion battery is used in this model due to the availability of data. The Li-ion battery specifications are 1 MW power rating and 5 MWh energy rating with approximately 4000 cycles [13]. In this simulation, only DC power is considered. Thus, converters and other power electronics issues and losses are ignored. Historical market data, including RMCCP, RMPCP, LMP, mileage ratio, performance score, and the RegD signal from PJM, are used to perform the simulation. The data utilized is from 12 days in 2019, a day from each month to represent an entire year [14].

The model is built using several binary variables to control market participation and make it easier to be linearized and coded in GAMS. We can choose which market the battery can participate in by adjusting equations (4) and (14)-(17) in the model. When bidding into the frequency regulation market, the battery must follow the RegD signal; otherwise, the performance score will decrease, which will significantly affect the payment over time. Participating in more than one market is more profitable than participating in just one market, but the significant point in the proposed model is bidding into both markets at the same time, if possible. In the first case, a new equation is applied for the battery to participate in one market (energy arbitrage OR frequency regulation up OR frequency regulation down) at each time interval as follows.

$$u^1_{td} + u^2_{td} + z^{rua}_{ktd} + z^{rda}_{ktd} \leq 1 \qquad \forall k, \forall t, \forall d \tag{25}$$

Thus, the BESS must choose the most profitable market, and it is possible that the BESS can buy from the energy market (charging the battery) and reserve it into the frequency regulation market (discharging the battery) in the next interval, or vice versa. In the first case, the simulation is run without limiting the daily cycles. The depth of discharge, in this case, is considered to be 90% of the battery's rated capacity. The total revenue of this case is \$552.72 for the two markets. The revenue generated from the energy arbitrage market is \$282.78 and from the frequency regulation market is \$296.93. The BESS performed 122 cycles during the 12 days.

In the second case, the model allows the battery to participate in both markets simultaneously and limits its daily cycles to 10. To compare it with the first case, we simply divided the number of cycles in the first case by the number of total days the BESS performed. The reasoning behind this is to have a fair comparison between cases, because we set unlimited cycles in the first case, which showed the maximum number of cycles the BESS can optimally perform while also achieving high revenue. Since participation in the frequency regulation market generates higher revenue, the energy arbitrage revenue is negative, because it is cheaper to buy from the energy market and to reserve or sell to the frequency regulation market. The energy arbitrage cost is \$458.23. Thus, the battery is buying more to follow the RegD signal. However, the total revenue is higher than in the first case. The total revenue is \$2,538.26. This case results in 65 cycles compared to 122 cycles in the first case, which will significantly affect battery performance and efficiency over time.

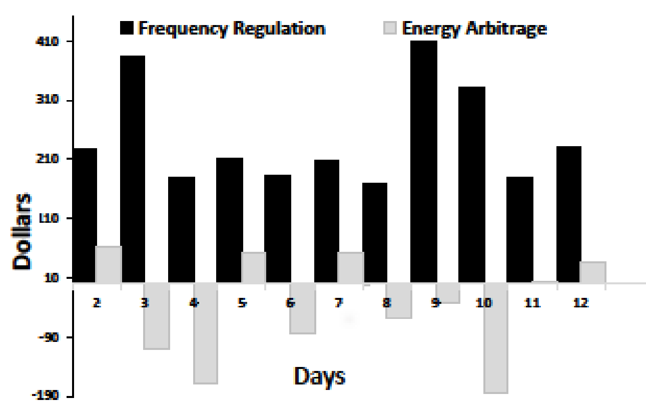

Fig. 2. Daily revenue for the second case.

V. CONCLUSION

A new MILP optimization model for a battery performing stacked services simultaneously is presented in this paper. Two applications are considered, energy arbitrage and frequency regulation, to evaluate a BESS's revenue using historical PJM market data. Two case studies are performed using the proposed model by modifying binary variables to adjust the model for market participation. The model aims to maximize profit by using optimal cycles and through strategies that involve bidding into multiple markets. Although only discharging cycles are considered, the second case achieves higher profits with fewer cycles. In the second case, the BESS buys more power from the grid for charging and discharges or reserves it for frequency regulation, which results in a negative revenue from the energy arbitrage market. Thus, the total revenue of the BESS increased by using the proposed scheduling method. Future work should examine the positive revenue impact over a longer term, such as over a year or more, to achieve a more comprehensive overview of BESS performance. Additionally, future studies should consider factors such as degradation and investment costs, and might also explore other applications of the proposed model to different areas such as voltage support and black-start. These additional examinations would provide BESS operators with an expanded and more accurate understanding of the long-term returns on investment of BESSs.

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